

TECHNICAL REPORT

Worldwide Disaster Trend Analysis

Technical documentation of the completed Power BI disaster analytics project

Document Control	Details
Project	Worldwide Disaster Analysis
Primary Analytical View	Worldwide Disaster Trend
Source Dataset	EM-DAT public dataset, supplied as Raw_Data.xlsx
Dashboard File	Worldwide Disaster Analysis.pbix
Raw Records	16,858
Raw Variables	47
Analysis Period	2000 to 2026
Source Version	2026-06-24
Report Type	Technical Data Analytics Report

Prepared from the supplied dashboard, Power BI project and raw workbook

1. Outline

This document follows a technical report structure for documenting the Worldwide Disaster Analysis project. It records the data source, data structure, preprocessing considerations, analytical methodology, dashboard outputs, technical validation, findings, limitations, recommendations and future analytical work.

1. Introduction
2. Story of Data
3. Data Splitting and Preprocessing
4. Pre-Analysis
5. In-Analysis
6. Post-Analysis and Insights
7. Data Visualizations and Charts
8. Recommendations and Observations
9. Conclusion
10. References and Appendices

2. Introduction

2.1 Project Objective

The objective is to analyse worldwide disaster occurrence and impact over time, with emphasis on disaster frequency, mortality, affected population and reported economic damage. The completed Power BI dashboard provides an interactive trend view with Country, Region, Disaster Type and Disaster Subtype filters.

2.2 Analytical Problem

The analysis converts a large historical disaster-event dataset into measurable indicators that identify peak disaster years, mortality peaks, population-impact peaks, economic-damage peaks and long-term event trends.

2.3 Scope

The supplied raw workbook contains 16,858 records and 47 variables. Start Year ranges from 2000 through 2026. The technical analysis focuses on the measures represented in the supplied Worldwide Disaster Trend dashboard and validates those outputs against the raw workbook.

2.4 Dashboard Architecture

Inspection of the supplied Power BI package shows report sections including Pre Analysis, Dashboard, In-Analysis 1, In-Analysis 2, In-Analysis 3, Post-Analysis and Recommendations. The supplied screenshot represents the Worldwide Disaster Trend analytical view.

2.5 Analytical Questions

- Which year recorded the highest number of disaster events?
- Which year recorded the highest total deaths?
- Which year recorded the highest total affected population?
- Which year recorded the highest reported total damage?
- How did disaster deaths, events, affected population and damage change over time?
- How can the dashboard be technically validated against the raw source data?

3. Story of Data

3.1 Data Source

The supplied EM-DAT Info sheet identifies the source as EM-DAT, CRED / UCLouvain, Brussels, Belgium. It records the public EM-DAT table version as 2026-06-24, file creation date as 29 June 2026, and 16,858 records.

Source Metadata	Value
Source	EM-DAT, CRED / UCLouvain, Brussels, Belgium
Table type	public_emdat
Version	2026-06-24
File creation	29 Jun 2026 02:47:31 UTC
Number of records	16,858

3.2 Dataset Structure

The raw table contains identifiers, disaster classification fields, geographic attributes, event dates, mortality measures, affected-population measures, financial impact measures and administrative metadata.

Category	Representative Variables
Identification	DisNo., External IDs, Event Name
Classification	Disaster Group, Disaster Subgroup, Disaster Type, Disaster Subtype
Geography	Country, ISO, Subregion, Region, Latitude, Longitude
Time	Start Year, Start Month, Start Day, End Year, End Month, End Day
Human impact	Total Deaths, No. Injured, No. Affected, No. Homeless, Total Affected
Economic impact	Reconstruction Costs, Insured Damage, Total Damage and adjusted variants
Metadata	CPI, Entry Date, Last Update, Admin Units, GADM Admin Units

3.3 Dataset Dimensions

Dimension	Result
Records	16,858
Variables	47
Countries	223
Regions	5
Disaster Types	31
Disaster Subtypes	63
Start Year Range	2000 to 2026
Duplicate DisNo. identifiers	0
Duplicate complete rows	0

3.4 Key Variables Used

Variable	Technical Role
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Start Year	Annual time dimension for trend aggregation.
DisNo.	Event identifier used to count disaster records.
Total Deaths	Annual mortality measure aggregated by Start Year.
Total Affected	Annual affected-population measure.
Total Damage ('000 US\$)	Reported damage measure expressed in thousands of US dollars.
Country	Geographic filter and segmentation dimension.
Region	Regional filter and segmentation dimension.
Disaster Type	Primary disaster classification filter.
Disaster Subtype	Detailed disaster classification filter.

4. Data Splitting and Preprocessing

4.1 Data Profiling

The raw workbook contains 16,858 records across 47 fields. No duplicated DisNo. identifiers and no exact duplicated complete rows were detected in the supplied workbook.

4.2 Missing-Value Assessment

Variable	Missing Records	Missing %
Total Deaths	3276	19.43%
Total Affected	4265	25.30%
Total Damage ('000 US\$)	13531	80.26%
Total Damage, Adjusted ('000 US\$)	13545	80.35%
Latitude	15004	89.00%
Longitude	15004	89.00%

The high missingness of economic and some impact fields is a material analytical consideration. Annual damage totals therefore represent reported values only and should not be interpreted as complete global economic loss.

4.3 Data Cleaning and Transformation Requirements

- Use Start Year as the annual time dimension.
- Count DisNo. records for disaster-event frequency.
- Aggregate Total Deaths and Total Affected by Start Year using sums of reported values.
- Aggregate Total Damage ('000 US\$) by Start Year using sums of source values.
- Retain Country, Region, Disaster Type and Disaster Subtype as dashboard filters.
- Preserve the distinction between unadjusted and adjusted financial damage.
- Do not treat missing financial values as zero unless explicitly defined by the analytical requirement.

4.4 Data Quality Considerations

- No duplicate event identifiers were detected.
- Impact and financial variables contain substantial missing values.
- The dataset contains both Natural and Technological disaster groups.
- 2026 is a partial year and should not be compared directly with complete historical years.

- The dashboard screenshot and raw workbook require synchronization checks because one KPI value does not agree with the raw-data aggregation and the damage KPI requires unit clarification.

5. Pre-Analysis

5.1 Descriptive Overview

Indicator	Raw Data Result
Highest event-count year	2000 (869 records)
Highest mortality year	2010 (336,900 deaths)
Highest affected-population year	2002 (659,481,311 affected)
Highest reported damage year	2011 (364,128,969 thousand US\$)

5.2 Top Disaster Types

Rank	Disaster Type	Records
1	Flood	4,273
2	Storm	2,877
3	Road	2,254
4	Water	1,187
5	Epidemic	894
6	Earthquake	698
7	Extreme temperature	568
8	Mass movement (wet)	500
9	Explosion (Industrial)	477
10	Fire (Miscellaneous)	461

5.3 Regional Distribution

Region	Records	Share
Asia	6,799	40.33%
Africa	4,327	25.67%
Americas	3,354	19.90%
Europe	1,957	11.61%
Oceania	421	2.50%

5.4 Initial Analytical Expectations

The pre-analysis indicates substantial variation by year, geography and disaster classification. Disaster frequency is expected to behave differently from severity measures because event count, mortality, population impact and economic damage represent different dimensions of disaster impact.

6. In-Analysis

6.1 Annual Disaster Event Trend

Using DisNo. as the event identifier, the raw data records a maximum of 869 events in 2000. The annual series fluctuates over the study period, with higher event counts concentrated in the early 2000s and further variation in later years.

6.2 Annual Mortality Trend

Total deaths peak in 2010 at 336,900. Other high-mortality years include 2004, 2008 and 2023. This demonstrates that annual mortality can be driven by a limited number of high-severity events.

6.3 Annual Affected-Population Trend

Total affected population peaks in 2002 at 659,481,311. The affected-population series is highly volatile, with major peaks in 2002 and 2015.

6.4 Annual Economic Damage Trend

The raw Total Damage ('000 US\$) field reaches its highest annual aggregate in 2011 at 364,128,969 thousand US\$.

Technical unit interpretation: 364,128,969 in a field measured in thousands of US dollars equals US\$364,128,969,000, approximately US\$364.129 billion. The dashboard KPI should therefore explicitly state its unit or convert the value before presentation.

6.5 Relationship Between Metrics

Event count measures frequency, deaths measure mortality, Total Affected measures human impact, and Total Damage measures reported financial loss. A peak in one metric should not be assumed to imply a peak in another.

6.6 Analytical Techniques

- Annual aggregation by Start Year
- Count aggregation for event frequency
- Sum aggregation for deaths, affected population and reported damage
- Peak-year ranking
- Geographic and categorical segmentation
- Time-series line charts
- KPI cards
- Dashboard-to-source reconciliation

7. Post-Analysis and Insights

7.1 Key Findings

Disaster frequency is highest in 2000. The raw dataset records 869 disaster records.

Mortality is highest in 2010. 2010 records 336,900 deaths.

Population impact peaks independently of mortality. 2002 records 659,481,311 total affected.

Economic damage peaks in 2011. The raw unadjusted damage aggregate is 364,128,969 thousand US\$.

Flood is the most frequent disaster type. Flood records total 4,273.

Asia has the largest regional event count. Asia records 6,799 events.

Reported damage is incomplete. 80.26% of records have no reported unadjusted damage value.

7.2 Technical Validation of Dashboard KPIs

Dashboard KPI	Dashboard Display	Raw Workbook Recalculation	Status
Peak Disaster Year	2000; 869 events	2000; 869 events	Validated
Peak Mortality Year	2010; 336,900 deaths	2010; 336,900 deaths	Validated
Year of Highest Damages	2011; displayed as \$364 Million	2011; 364,128,969 thousand US\$ = \$364.129 billion	Unit interpretation requires correction
Highest Affected Population Year	2002; displayed value 336,900	2002; 659,481,311 affected	KPI value mismatch

Technical control observation: the peak years for events, mortality, affected population and damage are supported by the raw workbook. The affected-population KPI displays 336,900 although the raw Total Affected aggregate for 2002 is 659,481,311. The damage KPI also requires explicit unit handling because the source field is denominated in thousands of US dollars.

7.3 Data-to-Dashboard Consistency

The supplied trend charts contain annual values that correspond to the source aggregation at visible points. The principal control issues identified from the supplied materials are concentrated in KPI presentation, unit labeling and synchronization between dashboard measures and source definitions.

8. Data Visualizations and Charts

8.1 Worldwide Disaster Trend Dashboard

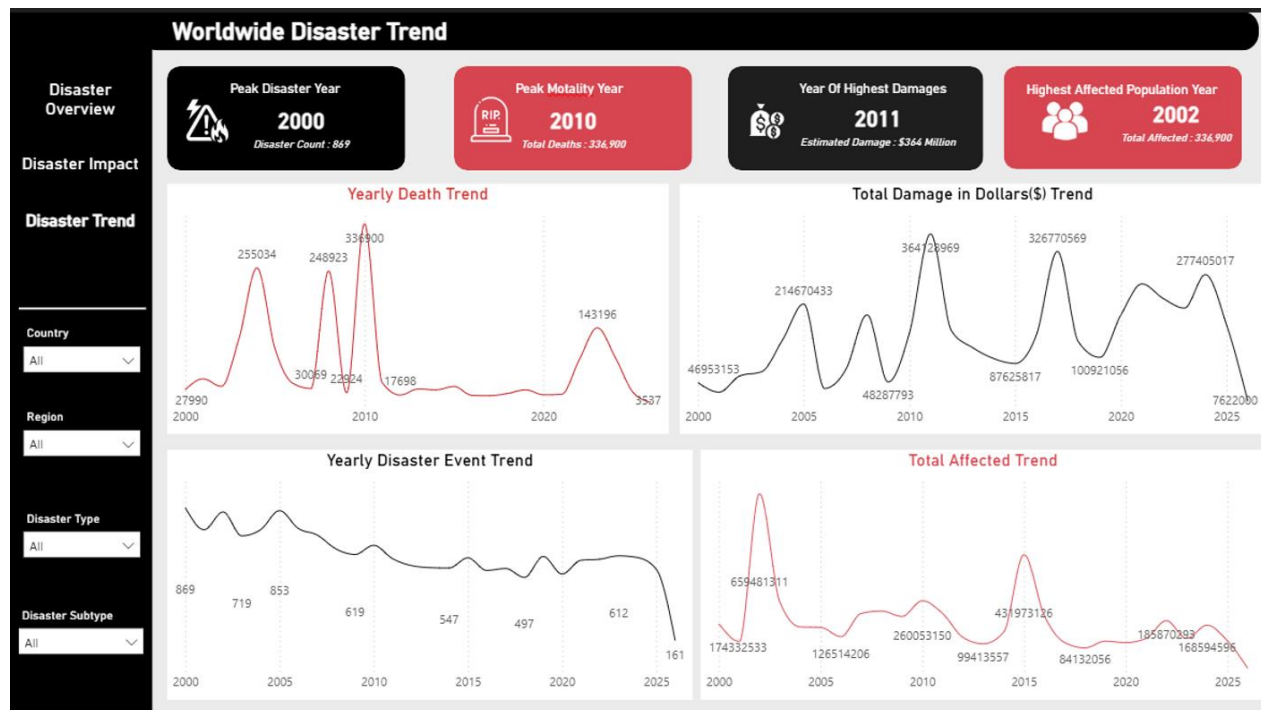


Figure 1. Supplied Power BI dashboard: Worldwide Disaster Trend.

The dashboard uses four KPI cards and four time-series charts. The KPI cards summarize peak event frequency, mortality, damage and affected population. The trend charts show annual deaths, damage, disaster events and affected population. Country, Region, Disaster Type and Disaster Subtype are interactive filters.

8.2 Visualization Technical Assessment

Visual	Purpose	Technical Assessment
KPI cards	Highlight peak years and headline totals	Effective for scanning; affected-population value and damage units require correction.
Yearly Death Trend	Show annual mortality volatility	Appropriate time-series visualization.
Total Damage Trend	Show annual economic impact	Appropriate if source unit is explicitly displayed.
Yearly Disaster Event Trend	Show annual event frequency	Appropriate for temporal frequency analysis.
Total Affected Trend	Show annual population impact	Appropriate, with a clear definition of Total Affected.
Slicers	Support geographic and disaster-category segmentation	Appropriate for interactive exploration.

8.3 Dashboard Design Observation

The visual hierarchy separates headline indicators from trend analysis and provides categorical filters on the left. For technical reporting, the dashboard should explicitly document measure definitions, source units, data freshness and completeness.

9. Recommendations and Observations

9.1 Technical Recommendations

Correct the affected-population KPI. Bind the KPI to the same Total Affected aggregation used by the trend measure so that 2002 displays 659,481,311 for the supplied workbook.

Correct damage units. Label the damage measure as thousand US\$ or convert it to million/billion US\$ before presentation.

Add data freshness context. State the EM-DAT version and extraction date, especially because 2026 is a partial year.

Document measure definitions. Define Event Count, Total Deaths, Total Affected and Total Damage in the dashboard.

Handle missing damage values explicitly. Do not interpret missing financial values as zero without a documented rule.

Add completeness indicators. Show the percentage of records with reported values for major impact metrics.

Add a partial-year warning. Prevent users from interpreting 2026 as a complete annual observation.

9.2 Analytical Recommendations

- Add disaster-severity metrics such as deaths per event and affected population per event.
- Add regional trend comparisons.
- Separate Natural and Technological disaster groups.
- Provide adjusted versus unadjusted damage views.

- Add a data-completeness panel.
- Introduce year-over-year percentage change for the core indicators.

9.3 Decision-Making Observation

The dashboard is suitable for descriptive disaster monitoring because it combines headline indicators, temporal trends and categorical filters. Before formal operational or policy use, data completeness, unit consistency and KPI reconciliation should be treated as mandatory controls.

10. Conclusion

10.1 Summary

The Worldwide Disaster Analysis project converts 16,858 EM-DAT records into an interactive temporal view of disaster frequency and impact. Raw-data validation confirms 2000 as the highest event-count year, 2010 as the highest mortality year, 2002 as the highest affected-population year, and 2011 as the highest reported unadjusted-damage year.

10.2 Technical Conclusion

The analytical structure is appropriate for descriptive disaster-trend analysis. The principal technical issues identified concern consistency of KPI presentation with the source data. The affected-population KPI requires correction, and the economic-damage KPI requires explicit unit handling.

10.3 Limitations

- The report is based on the supplied raw workbook, Power BI project and dashboard screenshot.
- The dataset contains substantial missing values in financial and some impact variables.
- The 2026 period is incomplete and should not be compared directly with complete historical years without qualification.
- No inferential statistical testing or causal analysis is evidenced in the supplied project materials.
- Reported economic damage represents recorded values and not necessarily total global economic loss.

10.4 Future Work

- Automate data-quality checks between Power BI and the raw source.
- Add anomaly detection for extreme disaster years.
- Develop regional and disaster-type drill-through pages.
- Create severity-normalized measures.
- Introduce adjusted-dollar analysis.
- Add a data dictionary and measure catalogue to the Power BI project.

11. References and Appendices

11.1 References

Source	Purpose
Raw_Data.xlsx, EM-DAT Data sheet	Record-level source for validation and aggregation.
Raw_Data.xlsx, EM-DAT Info sheet	Source metadata, version, table type and record count.
Worldwide Disaster Analysis.pbix	Power BI analytical model and dashboard implementation.
Supplied Worldwide Disaster Trend screenshot	Visual evidence of the completed dashboard and displayed KPI values.

EM-DAT documentation referenced in the source workbook	Data structure and glossary context.
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11.2 Appendix A: Core Annual Validation Metrics

Year	Events	Deaths	Total Affected	Total Damage ('000 US\$)
2000	869	27,990	174,332,533	46,953,153
2002	846	33,997	659,481,311	61,528,559
2005	853	105,158	160,831,210	214,670,433
2010	668	336,900	260,053,150	152,830,085
2011	597	40,774	212,841,342	364,128,969
2015	601	33,811	431,973,126	87,625,817
2020	515	18,195	103,798,797	191,247,447
2023	612	143,196	119,836,533	206,134,368
2024	603	88,271	168,594,596	277,405,017
2025	540	22,551	110,861,025	169,683,411
2026	161	3,537	7,759,383	7,622,000

11.3 Appendix B: Top Disaster Types by Record Count

Rank	Disaster Type	Records
1	Flood	4,273
2	Storm	2,877
3	Road	2,254
4	Water	1,187
5	Epidemic	894
6	Earthquake	698
7	Extreme temperature	568
8	Mass movement (wet)	500
9	Explosion (Industrial)	477
10	Fire (Miscellaneous)	461

11.4 Appendix C: Data Quality Summary

Quality Check	Result	Interpretation
Duplicate DisNo. identifiers	0	No duplicate event identifiers detected.
Duplicate complete rows	0	No exact duplicate rows detected.
Total Deaths missing	19.43%	Material missingness.
Total Affected missing	25.30%	Material missingness.
Total Damage missing	80.26%	High missingness; damage totals are incomplete.
2026 observation	Partial year	Do not compare directly with complete annual totals.

11.5 Appendix D: Technical Measure Definitions

Measure	Definition Used
Disaster Event Count	Count of disaster records identified by DisNo. grouped by Start Year.
Total Deaths	Sum of Total Deaths by Start Year for reported source values.
Total Affected	Sum of Total Affected by Start Year for reported source values.
Total Damage	Sum of Total Damage ('000 US\$) by Start Year; source units are thousands of US dollars.

11.6 Appendix E: Dashboard Evidence

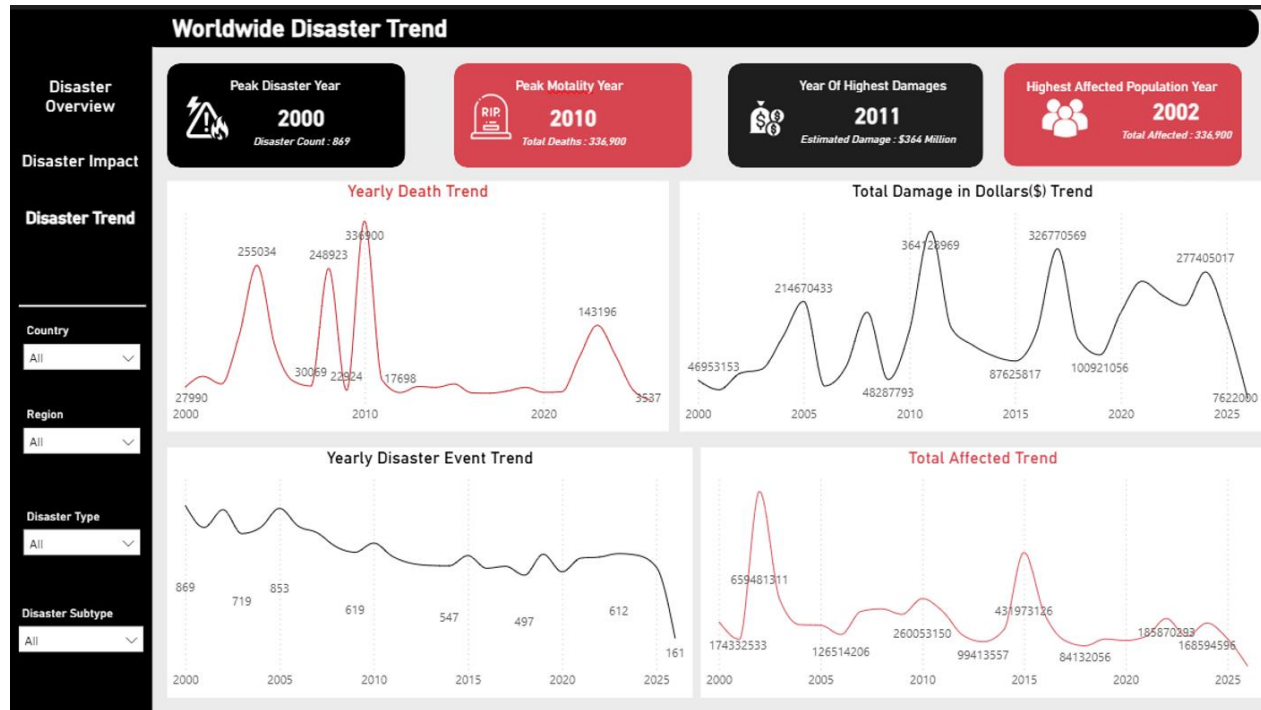


Figure 1. Worldwide Disaster Trend dashboard supplied with the project.